THESIS_PRIMARY_regression_equal_vs_shrinkage.csv

method	contrast	estimate	lower_Shrinkage	lower_Equal−variance	upper_Shrinkage	upper_Equal−variance	width_Shrinkage	width_Equal−variance
Regression MI	1 vs 0	0.257	−0.128	−0.134	0.643	0.649	0.772	0.782
Regression MI	2 vs 0	0.313	−0.051	−0.060	0.678	0.687	0.730	0.747
Regression MI	3 vs 0	0.541	0.114	0.111	0.968	0.970	0.854	0.859
Regression MI	4 vs 0	0.351	−0.071	−0.070	0.772	0.771	0.843	0.840

THESIS_PRIMARY_regression_equal_vs_shrinkage.csv

significant_Shrinkage	significant_Equal−variance
FALSE	FALSE
FALSE	FALSE
TRUE	TRUE
FALSE	FALSE

REFERENCE_METHOD_propensity_equal_vs_shrinkage.csv

method	contrast	estimate	lower_Shrinkage	lower_Equal−variance	upper_Shrinkage	upper_Equal−variance	width_Shrinkage	width_Equal−variance
Propensity MI	1 vs 0	0.265	−0.114	−0.117	0.643	0.646	0.757	0.763
Propensity MI	2 vs 0	0.321	−0.074	−0.079	0.716	0.722	0.790	0.801
Propensity MI	3 vs 0	0.538	0.127	0.126	0.950	0.951	0.823	0.825
Propensity MI	4 vs 0	0.388	−0.038	−0.040	0.814	0.815	0.852	0.855

REFERENCE_METHOD_propensity_equal_vs_shrinkage.csv

significant_Shrinkage	significant_Equal−variance
FALSE	FALSE
FALSE	FALSE
TRUE	TRUE
FALSE	FALSE

ALL_METHODS_full_precision_intervals.csv

method	contrast	estimate	lower	upper	width	significant	variance_method	shrink_alpha
Complete data	1 vs 0	0.285	−0.022	0.591	0.612	FALSE	Shrinkage	0.974
Complete data	2 vs 0	0.297	−0.011	0.605	0.617	FALSE	Shrinkage	0.974
Complete data	3 vs 0	0.351	0.039	0.663	0.624	TRUE	Shrinkage	0.974
Complete data	4 vs 0	0.348	0.037	0.659	0.622	TRUE	Shrinkage	0.974
Complete data	1 vs 0	0.285	−0.023	0.592	0.615	FALSE	Non−shrinkage	NA
Complete data	2 vs 0	0.297	0.015	0.579	0.564	TRUE	Non−shrinkage	NA
Complete data	3 vs 0	0.351	0.049	0.653	0.604	TRUE	Non−shrinkage	NA
Complete data	4 vs 0	0.348	0.038	0.658	0.620	TRUE	Non−shrinkage	NA
Regression MI	1 vs 0	0.257	−0.128	0.643	0.772	FALSE	Shrinkage	0.714
Regression MI	2 vs 0	0.313	−0.051	0.678	0.730	FALSE	Shrinkage	0.714
Regression MI	3 vs 0	0.541	0.114	0.968	0.854	TRUE	Shrinkage	0.714
Regression MI	4 vs 0	0.351	−0.071	0.772	0.843	FALSE	Shrinkage	0.714
Regression MI	1 vs 0	0.257	−0.119	0.634	0.752	FALSE	Non−shrinkage	NA
Regression MI	2 vs 0	0.313	−0.036	0.662	0.698	FALSE	Non−shrinkage	NA
Regression MI	3 vs 0	0.541	0.117	0.964	0.847	TRUE	Non−shrinkage	NA
Regression MI	4 vs 0	0.351	−0.071	0.772	0.843	FALSE	Non−shrinkage	NA
Regression MI	1 vs 0	0.257	−0.134	0.649	0.782	FALSE	Equal−variance	NA
Regression MI	2 vs 0	0.313	−0.060	0.687	0.747	FALSE	Equal−variance	NA
Regression MI	3 vs 0	0.541	0.111	0.970	0.859	TRUE	Equal−variance	NA
Regression MI	4 vs 0	0.351	−0.070	0.771	0.840	FALSE	Equal−variance	NA

ALL_METHODS_full_precision_intervals.csv

method	contrast	estimate	lower	upper	width	significant	variance_method	shrink_alpha
PMM MI	1 vs 0	0.288	−0.056	0.632	0.688	FALSE	Shrinkage	0.751
PMM MI	2 vs 0	0.334	−0.007	0.676	0.683	FALSE	Shrinkage	0.751
PMM MI	3 vs 0	0.503	0.106	0.900	0.794	TRUE	Shrinkage	0.751
PMM MI	4 vs 0	0.351	−0.016	0.718	0.734	FALSE	Shrinkage	0.751
PMM MI	1 vs 0	0.288	−0.045	0.621	0.666	FALSE	Non−shrinkage	NA
PMM MI	2 vs 0	0.334	0.009	0.660	0.650	TRUE	Non−shrinkage	NA
PMM MI	3 vs 0	0.503	0.112	0.894	0.782	TRUE	Non−shrinkage	NA
PMM MI	4 vs 0	0.351	−0.016	0.718	0.734	FALSE	Non−shrinkage	NA
PMM MI	1 vs 0	0.288	−0.061	0.637	0.698	FALSE	Equal−variance	NA
PMM MI	2 vs 0	0.334	−0.014	0.683	0.696	FALSE	Equal−variance	NA
PMM MI	3 vs 0	0.503	0.104	0.902	0.798	TRUE	Equal−variance	NA
PMM MI	4 vs 0	0.351	−0.016	0.718	0.734	FALSE	Equal−variance	NA
Propensity MI	1 vs 0	0.265	−0.114	0.643	0.757	FALSE	Shrinkage	0.774
Propensity MI	2 vs 0	0.321	−0.074	0.716	0.790	FALSE	Shrinkage	0.774
Propensity MI	3 vs 0	0.538	0.127	0.950	0.823	TRUE	Shrinkage	0.774
Propensity MI	4 vs 0	0.388	−0.038	0.814	0.852	FALSE	Shrinkage	0.774
Propensity MI	1 vs 0	0.265	−0.106	0.635	0.741	FALSE	Non−shrinkage	NA
Propensity MI	2 vs 0	0.321	−0.061	0.703	0.764	FALSE	Non−shrinkage	NA
Propensity MI	3 vs 0	0.538	0.133	0.944	0.811	TRUE	Non−shrinkage	NA
Propensity MI	4 vs 0	0.388	−0.036	0.811	0.848	FALSE	Non−shrinkage	NA

ALL_METHODS_full_precision_intervals.csv

method	contrast	estimate	lower	upper	width	significant	variance_method	shrink_alpha
Propensity MI	1 vs 0	0.265	−0.117	0.646	0.763	FALSE	Equal−variance	NA
Propensity MI	2 vs 0	0.321	−0.079	0.722	0.801	FALSE	Equal−variance	NA
Propensity MI	3 vs 0	0.538	0.126	0.951	0.825	TRUE	Equal−variance	NA
Propensity MI	4 vs 0	0.388	−0.040	0.815	0.855	FALSE	Equal−variance	NA

completion_summary.csv

analysis	imputations
checkpoints/pmm_20461107	40
checkpoints/pmm_22381107	40
checkpoints/pmm_22481107	40
checkpoints/pmm_22581107	40
checkpoints/propensity_20561107	40
checkpoints/propensity_22391107	40
checkpoints/propensity_22491107	40
checkpoints/propensity_22591107	40
checkpoints/regression_20361107	40
checkpoints/regression_22371107	40
checkpoints/regression_22471107	40
checkpoints/regression_22571107	40

APPENDIX_10_20_40pct_MAR_sensitivity.csv

target_missing_rate	achieved_missing_rate	method	contrast	estimate	lower	upper	width	significant
0.1	0.117	Regression MI	1 vs 0	0.282	−0.037	0.601	0.638	FALSE
0.1	0.117	Regression MI	2 vs 0	0.296	−0.018	0.610	0.628	FALSE
0.1	0.117	Regression MI	3 vs 0	0.273	−0.047	0.593	0.640	FALSE
0.1	0.117	Regression MI	4 vs 0	0.291	−0.043	0.625	0.668	FALSE
0.1	0.117	Regression MI	1 vs 0	0.282	−0.039	0.604	0.644	FALSE
0.1	0.117	Regression MI	2 vs 0	0.296	−0.003	0.595	0.599	FALSE
0.1	0.117	Regression MI	3 vs 0	0.273	−0.033	0.578	0.611	FALSE
0.1	0.117	Regression MI	4 vs 0	0.291	−0.043	0.625	0.668	FALSE
0.1	0.117	Regression MI	1 vs 0	0.282	−0.035	0.600	0.636	FALSE
0.1	0.117	Regression MI	2 vs 0	0.296	−0.027	0.619	0.646	FALSE
0.1	0.117	Regression MI	3 vs 0	0.273	−0.056	0.602	0.658	FALSE
0.1	0.117	Regression MI	4 vs 0	0.291	−0.044	0.625	0.669	FALSE
0.1	0.117	PMM MI	1 vs 0	0.298	−0.021	0.617	0.639	FALSE
0.1	0.117	PMM MI	2 vs 0	0.292	−0.017	0.601	0.617	FALSE
0.1	0.117	PMM MI	3 vs 0	0.274	−0.041	0.589	0.630	FALSE
0.1	0.117	PMM MI	4 vs 0	0.309	−0.012	0.629	0.641	FALSE
0.1	0.117	PMM MI	1 vs 0	0.298	−0.024	0.620	0.644	FALSE
0.1	0.117	PMM MI	2 vs 0	0.292	−0.001	0.585	0.585	FALSE
0.1	0.117	PMM MI	3 vs 0	0.274	−0.027	0.576	0.603	FALSE
0.1	0.117	PMM MI	4 vs 0	0.309	−0.012	0.629	0.640	FALSE

APPENDIX_10_20_40pct_MAR_sensitivity.csv

target_missing_rate	achieved_missing_rate	method	contrast	estimate	lower	upper	width	significant
0.1	0.117	PMM MI	1 vs 0	0.298	−0.020	0.616	0.637	FALSE
0.1	0.117	PMM MI	2 vs 0	0.292	−0.027	0.611	0.638	FALSE
0.1	0.117	PMM MI	3 vs 0	0.274	−0.049	0.598	0.648	FALSE
0.1	0.117	PMM MI	4 vs 0	0.309	−0.012	0.629	0.642	FALSE
0.1	0.117	Propensity MI	1 vs 0	0.283	−0.044	0.609	0.653	FALSE
0.1	0.117	Propensity MI	2 vs 0	0.282	−0.042	0.607	0.649	FALSE
0.1	0.117	Propensity MI	3 vs 0	0.269	−0.052	0.591	0.644	FALSE
0.1	0.117	Propensity MI	4 vs 0	0.299	−0.032	0.629	0.661	FALSE
0.1	0.117	Propensity MI	1 vs 0	0.283	−0.047	0.613	0.660	FALSE
0.1	0.117	Propensity MI	2 vs 0	0.282	−0.028	0.592	0.620	FALSE
0.1	0.117	Propensity MI	3 vs 0	0.269	−0.038	0.577	0.615	FALSE
0.1	0.117	Propensity MI	4 vs 0	0.299	−0.031	0.629	0.660	FALSE
0.1	0.117	Propensity MI	1 vs 0	0.283	−0.043	0.608	0.651	FALSE
0.1	0.117	Propensity MI	2 vs 0	0.282	−0.050	0.615	0.664	FALSE
0.1	0.117	Propensity MI	3 vs 0	0.269	−0.060	0.599	0.659	FALSE
0.1	0.117	Propensity MI	4 vs 0	0.299	−0.032	0.629	0.661	FALSE
0.2	0.163	Regression MI	1 vs 0	0.293	−0.038	0.625	0.663	FALSE
0.2	0.163	Regression MI	2 vs 0	0.323	0.003	0.642	0.639	TRUE
0.2	0.163	Regression MI	3 vs 0	0.390	0.045	0.735	0.690	TRUE
0.2	0.163	Regression MI	4 vs 0	0.358	0.026	0.690	0.664	TRUE

APPENDIX_10_20_40pct_MAR_sensitivity.csv

target_missing_rate	achieved_missing_rate	method	contrast	estimate	lower	upper	width	significant
0.2	0.163	Regression MI	1 vs 0	0.293	−0.035	0.621	0.656	FALSE
0.2	0.163	Regression MI	2 vs 0	0.323	0.024	0.621	0.597	TRUE
0.2	0.163	Regression MI	3 vs 0	0.390	0.049	0.731	0.682	TRUE
0.2	0.163	Regression MI	4 vs 0	0.358	0.022	0.693	0.672	TRUE
0.2	0.163	Regression MI	1 vs 0	0.293	−0.039	0.626	0.666	FALSE
0.2	0.163	Regression MI	2 vs 0	0.323	−0.003	0.648	0.652	FALSE
0.2	0.163	Regression MI	3 vs 0	0.390	0.044	0.736	0.692	TRUE
0.2	0.163	Regression MI	4 vs 0	0.358	0.027	0.688	0.661	TRUE
0.2	0.163	PMM MI	1 vs 0	0.301	−0.023	0.624	0.648	FALSE
0.2	0.163	PMM MI	2 vs 0	0.295	−0.030	0.620	0.650	FALSE
0.2	0.163	PMM MI	3 vs 0	0.363	0.030	0.697	0.667	TRUE
0.2	0.163	PMM MI	4 vs 0	0.347	0.012	0.681	0.669	TRUE
0.2	0.163	PMM MI	1 vs 0	0.301	−0.020	0.621	0.641	FALSE
0.2	0.163	PMM MI	2 vs 0	0.295	−0.010	0.600	0.610	FALSE
0.2	0.163	PMM MI	3 vs 0	0.363	0.035	0.691	0.656	TRUE
0.2	0.163	PMM MI	4 vs 0	0.347	0.009	0.685	0.676	TRUE
0.2	0.163	PMM MI	1 vs 0	0.301	−0.024	0.626	0.650	FALSE
0.2	0.163	PMM MI	2 vs 0	0.295	−0.038	0.627	0.665	FALSE
0.2	0.163	PMM MI	3 vs 0	0.363	0.028	0.698	0.670	TRUE
0.2	0.163	PMM MI	4 vs 0	0.347	0.014	0.680	0.666	TRUE

APPENDIX_10_20_40pct_MAR_sensitivity.csv

target_missing_rate	achieved_missing_rate	method	contrast	estimate	lower	upper	width	significant
0.2	0.163	Propensity MI	1 vs 0	0.297	−0.037	0.631	0.669	FALSE
0.2	0.163	Propensity MI	2 vs 0	0.307	−0.017	0.631	0.647	FALSE
0.2	0.163	Propensity MI	3 vs 0	0.390	0.045	0.736	0.691	TRUE
0.2	0.163	Propensity MI	4 vs 0	0.366	0.020	0.711	0.690	TRUE
0.2	0.163	Propensity MI	1 vs 0	0.297	−0.034	0.628	0.663	FALSE
0.2	0.163	Propensity MI	2 vs 0	0.307	0.003	0.611	0.608	TRUE
0.2	0.163	Propensity MI	3 vs 0	0.390	0.049	0.732	0.683	TRUE
0.2	0.163	Propensity MI	4 vs 0	0.366	0.018	0.714	0.696	TRUE
0.2	0.163	Propensity MI	1 vs 0	0.297	−0.039	0.633	0.672	FALSE
0.2	0.163	Propensity MI	2 vs 0	0.307	−0.024	0.638	0.662	FALSE
0.2	0.163	Propensity MI	3 vs 0	0.390	0.044	0.737	0.693	TRUE
0.2	0.163	Propensity MI	4 vs 0	0.366	0.021	0.710	0.689	TRUE
0.4	0.396	Regression MI	1 vs 0	0.263	−0.159	0.686	0.845	FALSE
0.4	0.396	Regression MI	2 vs 0	0.315	−0.074	0.704	0.778	FALSE
0.4	0.396	Regression MI	3 vs 0	0.312	−0.058	0.682	0.740	FALSE
0.4	0.396	Regression MI	4 vs 0	0.521	0.061	0.980	0.919	TRUE
0.4	0.396	Regression MI	1 vs 0	0.263	−0.160	0.686	0.846	FALSE
0.4	0.396	Regression MI	2 vs 0	0.315	−0.063	0.693	0.756	FALSE
0.4	0.396	Regression MI	3 vs 0	0.312	−0.050	0.675	0.725	FALSE
0.4	0.396	Regression MI	4 vs 0	0.521	0.058	0.984	0.926	TRUE

APPENDIX_10_20_40pct_MAR_sensitivity.csv

target_missing_rate	achieved_missing_rate	method	contrast	estimate	lower	upper	width	significant
0.4	0.396	Regression MI	1 vs 0	0.263	−0.159	0.685	0.844	FALSE
0.4	0.396	Regression MI	2 vs 0	0.315	−0.086	0.716	0.801	FALSE
0.4	0.396	Regression MI	3 vs 0	0.312	−0.067	0.692	0.759	FALSE
0.4	0.396	Regression MI	4 vs 0	0.521	0.066	0.975	0.909	TRUE
0.4	0.396	PMM MI	1 vs 0	0.281	−0.127	0.689	0.816	FALSE
0.4	0.396	PMM MI	2 vs 0	0.331	−0.012	0.674	0.686	FALSE
0.4	0.396	PMM MI	3 vs 0	0.320	−0.046	0.687	0.733	FALSE
0.4	0.396	PMM MI	4 vs 0	0.459	0.051	0.868	0.817	TRUE
0.4	0.396	PMM MI	1 vs 0	0.281	−0.131	0.693	0.823	FALSE
0.4	0.396	PMM MI	2 vs 0	0.331	−0.001	0.663	0.664	FALSE
0.4	0.396	PMM MI	3 vs 0	0.320	−0.038	0.679	0.717	FALSE
0.4	0.396	PMM MI	4 vs 0	0.459	0.048	0.871	0.824	TRUE
0.4	0.396	PMM MI	1 vs 0	0.281	−0.124	0.686	0.811	FALSE
0.4	0.396	PMM MI	2 vs 0	0.331	−0.021	0.683	0.703	FALSE
0.4	0.396	PMM MI	3 vs 0	0.320	−0.053	0.694	0.747	FALSE
0.4	0.396	PMM MI	4 vs 0	0.459	0.053	0.865	0.812	TRUE
0.4	0.396	Propensity MI	1 vs 0	0.276	−0.154	0.706	0.860	FALSE
0.4	0.396	Propensity MI	2 vs 0	0.334	−0.066	0.733	0.799	FALSE
0.4	0.396	Propensity MI	3 vs 0	0.342	−0.046	0.729	0.775	FALSE
0.4	0.396	Propensity MI	4 vs 0	0.483	0.055	0.911	0.856	TRUE

APPENDIX_10_20_40pct_MAR_sensitivity.csv

target_missing_rate	achieved_missing_rate	method	contrast	estimate	lower	upper	width	significant
0.4	0.396	Propensity MI	1 vs 0	0.276	−0.157	0.710	0.867	FALSE
0.4	0.396	Propensity MI	2 vs 0	0.334	−0.057	0.724	0.781	FALSE
0.4	0.396	Propensity MI	3 vs 0	0.342	−0.039	0.723	0.761	FALSE
0.4	0.396	Propensity MI	4 vs 0	0.483	0.050	0.917	0.866	TRUE
0.4	0.396	Propensity MI	1 vs 0	0.276	−0.151	0.704	0.855	FALSE
0.4	0.396	Propensity MI	2 vs 0	0.334	−0.074	0.741	0.815	FALSE
0.4	0.396	Propensity MI	3 vs 0	0.342	−0.053	0.737	0.790	FALSE
0.4	0.396	Propensity MI	4 vs 0	0.483	0.060	0.907	0.848	TRUE

APPENDIX_10_20_40pct_MAR_sensitivity.csv

variance_method	shrink_alpha
Shrinkage	0.644
Shrinkage	0.644
Shrinkage	0.644
Shrinkage	0.644
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Shrinkage	0.624
Shrinkage	0.624
Shrinkage	0.624
Shrinkage	0.624
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA

APPENDIX_10_20_40pct_MAR_sensitivity.csv

variance_method	shrink_alpha
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Shrinkage	0.680
Shrinkage	0.680
Shrinkage	0.680
Shrinkage	0.680
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Shrinkage	0.793
Shrinkage	0.793
Shrinkage	0.793
Shrinkage	0.793

APPENDIX_10_20_40pct_MAR_sensitivity.csv

variance_method	shrink_alpha
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Shrinkage	0.754
Shrinkage	0.754
Shrinkage	0.754
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA

APPENDIX_10_20_40pct_MAR_sensitivity.csv

variance_method	shrink_alpha
Shrinkage	0.773
Shrinkage	0.773
Shrinkage	0.773
Shrinkage	0.773
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Shrinkage	0.572
Shrinkage	0.572
Shrinkage	0.572
Shrinkage	0.572
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA

APPENDIX_10_20_40pct_MAR_sensitivity.csv

variance_method	shrink_alpha
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Shrinkage	0.630
Shrinkage	0.630
Shrinkage	0.630
Shrinkage	0.630
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Shrinkage	0.606
Shrinkage	0.606
Shrinkage	0.606
Shrinkage	0.606

APPENDIX_10_20_40pct_MAR_sensitivity.csv

variance_method	shrink_alpha
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Non−shrinkage	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA
Equal−variance	NA

APPENDIX_group_summary.csv

group	n	mean	sd	variance
0	71	0.217	0.695	0.483
1	78	0.502	0.830	0.689
2	75	0.514	0.690	0.476
3	72	0.568	0.771	0.595
4	73	0.565	0.812	0.660

APPENDIX_primary_missingness_by_arm.csv

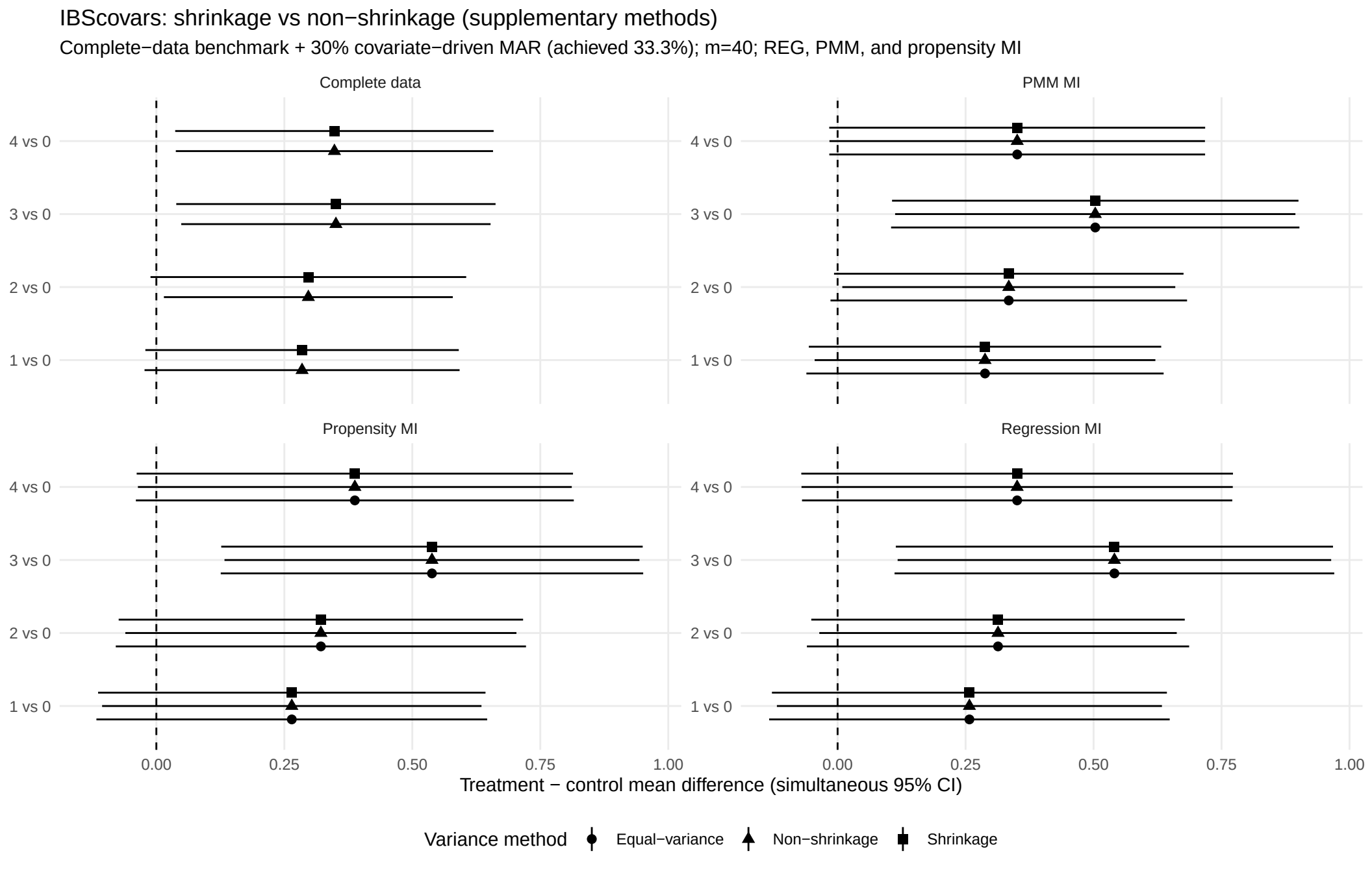
group	n	n_missing	missing_pct
0	71	18	25.352
1	78	28	35.897
2	75	24	32.000
3	72	24	33.333
4	73	29	39.726

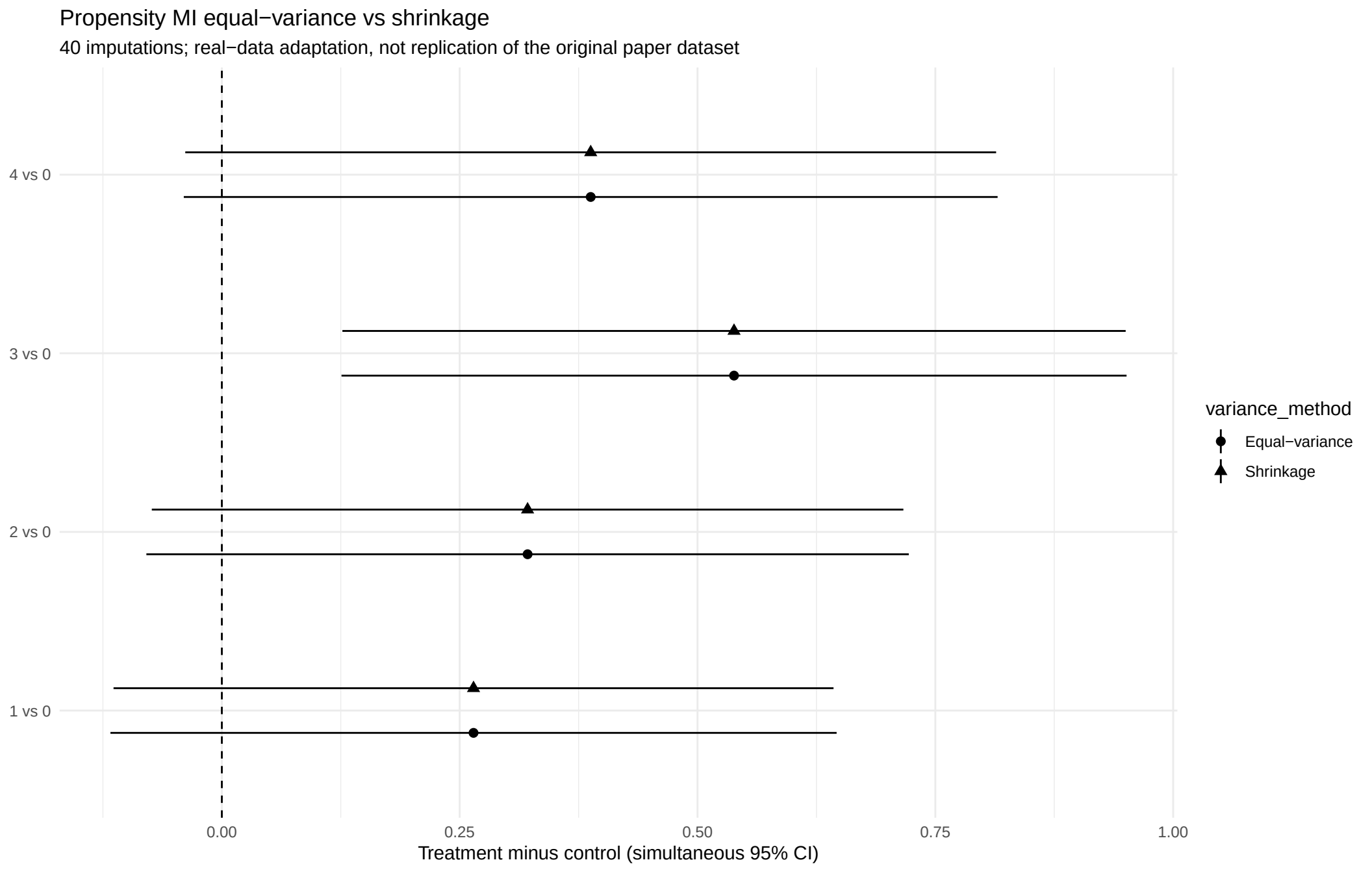
APPENDIX_shrinkage_alpha.csv

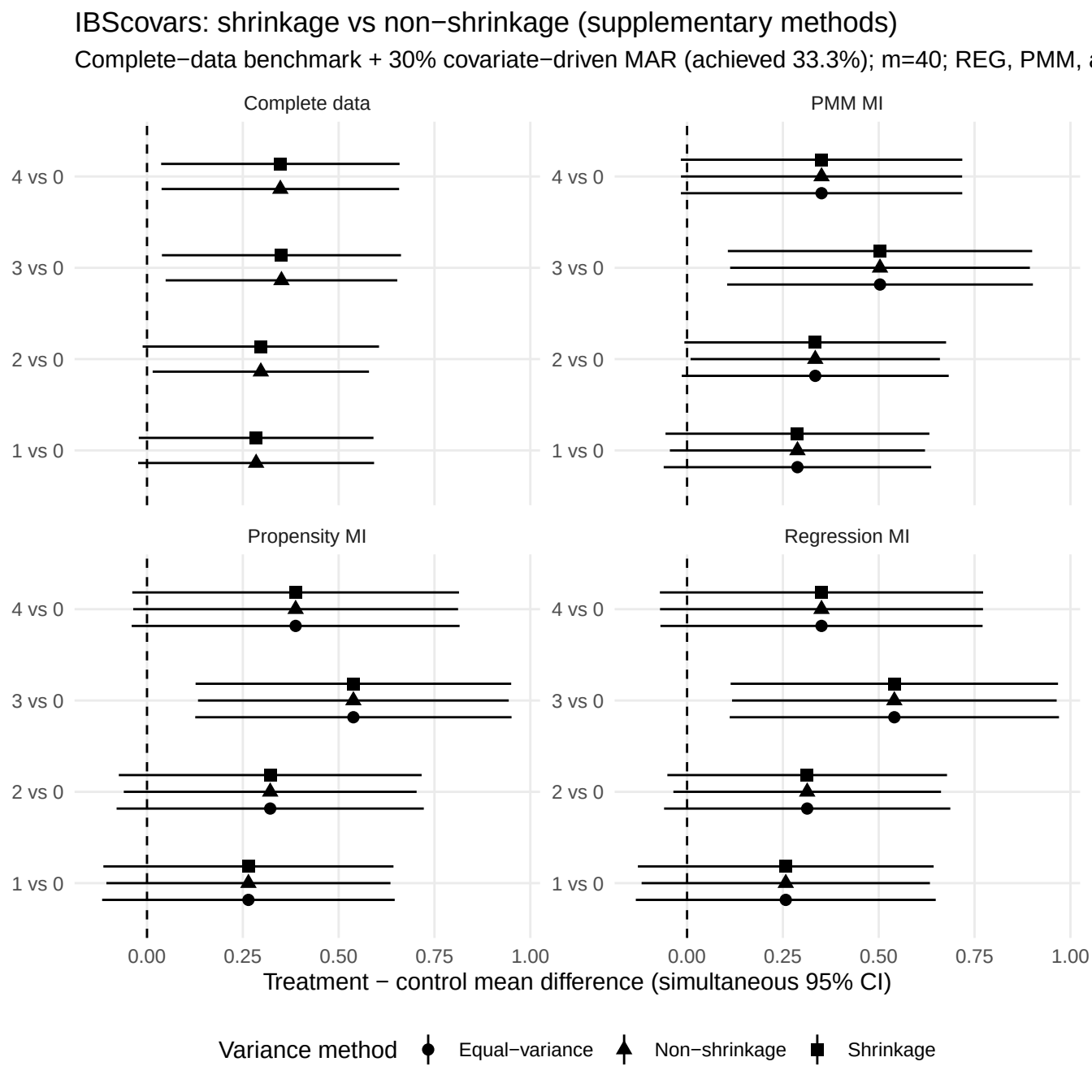
method	contrast	shrink_alpha
Complete data	1 vs 0	0.974
Complete data	2 vs 0	0.974
Complete data	3 vs 0	0.974
Complete data	4 vs 0	0.974
Regression MI	1 vs 0	0.714
Regression MI	2 vs 0	0.714
Regression MI	3 vs 0	0.714
Regression MI	4 vs 0	0.714
PMM MI	1 vs 0	0.751
PMM MI	2 vs 0	0.751
PMM MI	3 vs 0	0.751
PMM MI	4 vs 0	0.751
Propensity MI	1 vs 0	0.774
Propensity MI	2 vs 0	0.774
Propensity MI	3 vs 0	0.774
Propensity MI	4 vs 0	0.774

IBScovars_SUPPLEMENTARY_PRIMARY_RESULT_TABLE.csv

Method	Contrast	Estimate	Shrinkage 95% CI	Non−shrinkage 95% CI	Width Sh	Width NS	Conclusion (Sh / NS)
Complete data	1 vs 0	0.285	[−0.022, 0.591]	[−0.023, 0.592]	0.612	0.615	Not significant / Not significant
Complete data	2 vs 0	0.297	[−0.011, 0.605]	[0.015, 0.579]	0.617	0.564	Not significant / Significant
Complete data	3 vs 0	0.351	[0.039, 0.663]	[0.049, 0.653]	0.624	0.604	Significant / Significant
Complete data	4 vs 0	0.348	[0.037, 0.659]	[0.038, 0.658]	0.622	0.620	Significant / Significant
Regression MI	1 vs 0	0.257	[−0.128, 0.643]	[−0.119, 0.634]	0.772	0.752	Not significant / Not significant
Regression MI	2 vs 0	0.313	[−0.051, 0.678]	[−0.036, 0.662]	0.730	0.698	Not significant / Not significant
Regression MI	3 vs 0	0.541	[0.114, 0.968]	[0.117, 0.964]	0.854	0.847	Significant / Significant
Regression MI	4 vs 0	0.351	[−0.071, 0.772]	[−0.071, 0.772]	0.843	0.843	Not significant / Not significant
PMM MI	1 vs 0	0.288	[−0.056, 0.632]	[−0.045, 0.621]	0.688	0.666	Not significant / Not significant
PMM MI	2 vs 0	0.334	[−0.007, 0.676]	[0.009, 0.660]	0.683	0.650	Not significant / Significant
PMM MI	3 vs 0	0.503	[0.106, 0.900]	[0.112, 0.894]	0.794	0.782	Significant / Significant
PMM MI	4 vs 0	0.351	[−0.016, 0.718]	[−0.016, 0.718]	0.734	0.734	Not significant / Not significant
Propensity MI	1 vs 0	0.265	[−0.114, 0.643]	[−0.106, 0.635]	0.757	0.741	Not significant / Not significant
Propensity MI	2 vs 0	0.321	[−0.074, 0.716]	[−0.061, 0.703]	0.790	0.764	Not significant / Not significant
Propensity MI	3 vs 0	0.538	[0.127, 0.950]	[0.133, 0.944]	0.823	0.811	Significant / Significant
Propensity MI	4 vs 0	0.388	[−0.038, 0.814]	[−0.036, 0.811]	0.852	0.848	Not significant / Not significant


IBScovars: shrinkage vs non−shrinkage (supplementary methods)
Complete−data benchmark + 30% covariate−driven MAR (achieved 33.3%); m=40; REG, PMM, and propensity MI
Complete data
PMM MI
Propensity MI
Regression MI
4 vs 0
3 vs 0
2 vs 0
1 vs 0
0.00
0.25
0.50
0.75
1.00
Treatment − control mean difference (simultaneous 95% CI)
Variance method
Equal−variance
Non−shrinkage
Shrinkage


Propensity MI equal−variance vs shrinkage
40 imputations; real−data adaptation, not replication of the original paper dataset
4 vs 0
3 vs 0
2 vs 0
1 vs 0
0.00
0.25
0.50
0.75
1.00
Treatment minus control (simultaneous 95% CI)
variance_method
Equal−variance
Shrinkage


IBScovars: shrinkage vs non−shrinkage (supplementary methods)
Complete−data benchmark + 30% covariate−driven MAR (achieved 33.3%); m=40; REG, PMM,
Complete data
PMM MI
Propensity MI
Regression MI
4 vs 0
3 vs 0
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1 vs 0
0.00
0.25
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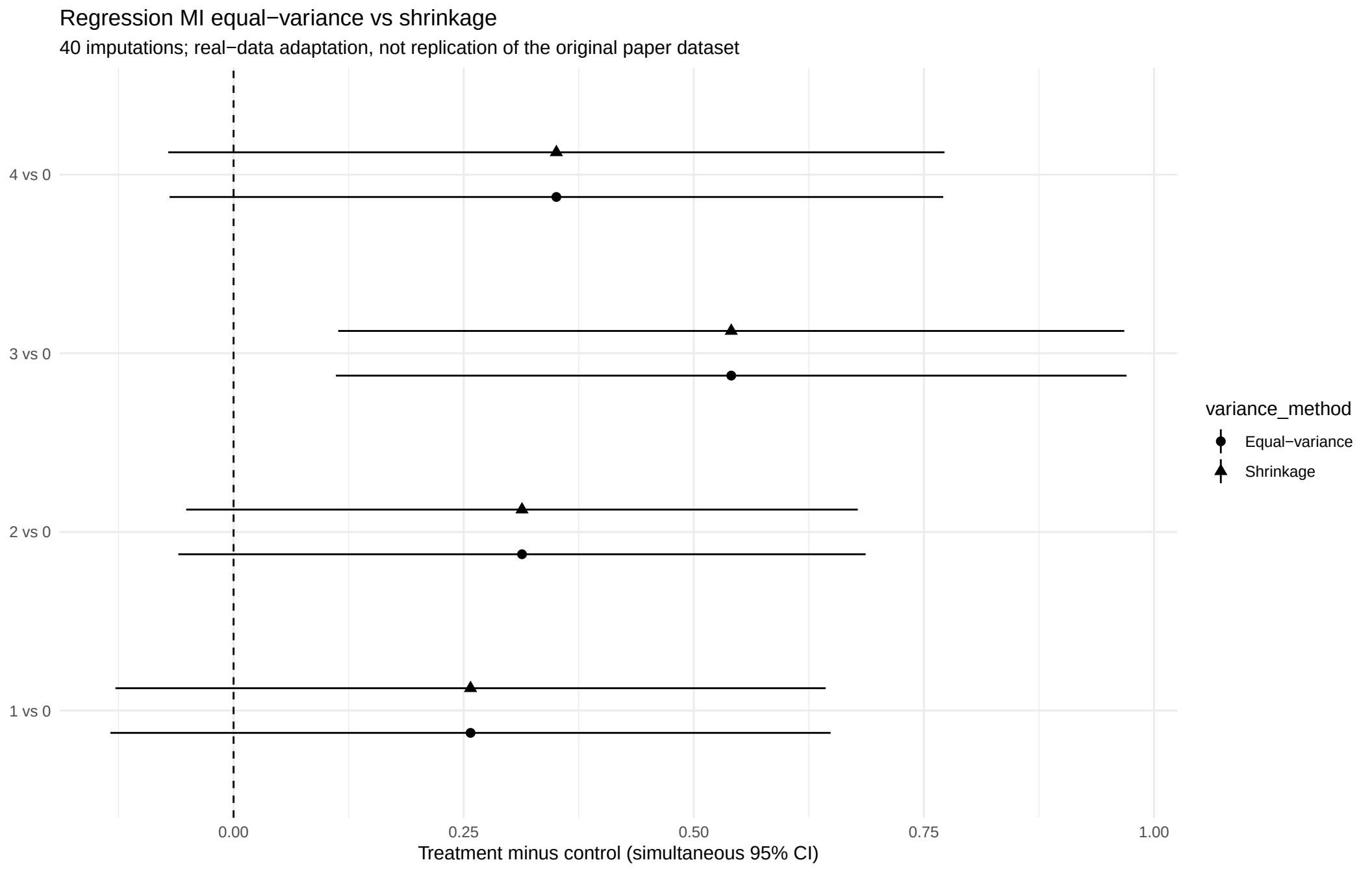
0.75
1.00
Treatment − control mean difference (simultaneous 95% CI)
Variance method
Equal−variance
Non−shrinkage
Shrinkage


Regression MI equal−variance vs shrinkage
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Treatment minus control (simultaneous 95% CI)
variance_method
Equal−variance
Shrinkage